Sequences & Series

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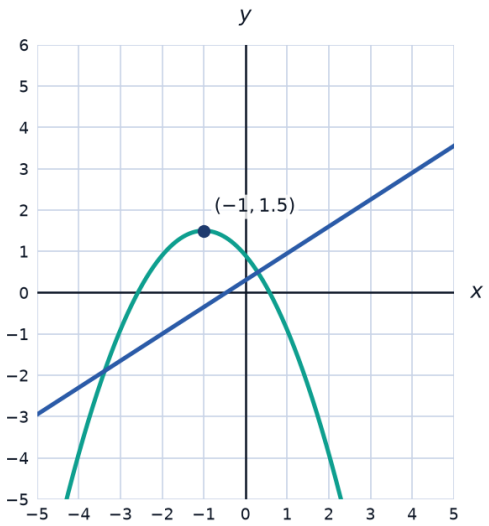
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Arithmetic sequences**

For the arithmetic sequence with $a_1 = 9$ and common difference $d = 6$, find a_{10}.

2 **[Foundational]** • **Target: Arithmetic sequences**

For the arithmetic sequence with $a_1 = 15$ and $d = -4$, find a_{11}.



3 **[Foundational]** • **Target: Arithmetic series**

Find S_{10} for the arithmetic sequence with $a_1 = 9$, $d = 6$ (using $a_{10} = 63$ from above).

4 **[Foundational]** • **Target: Geometric sequences**

For the geometric sequence with $a_1 = 6$ and common ratio $r = 2$, find a_8.

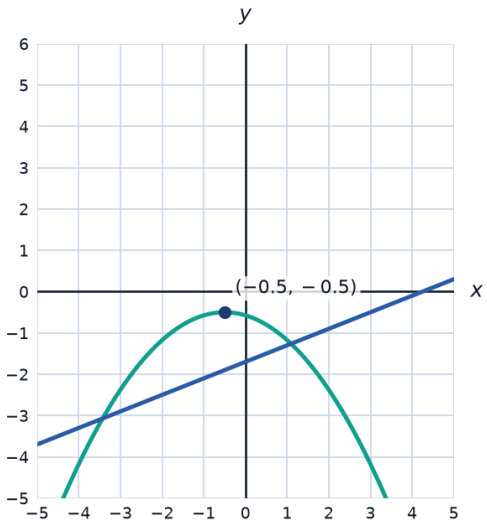
5 **[Foundational]** • **Target: Geometric sequences**

A geometric sequence has $a_1 = 2$ and common ratio $r = 5$. Find the smallest n for which $a_n = 1250$.

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6 [Foundational] • Target: Geometric series (finite)

Find S_8 for the geometric sequence with $a_1 = 6$, $r = 2$ (using $a_8 = 768$ from above).



7 [Foundational] • Target: Infinite geometric series

Find the sum of the infinite geometric series $36 + 12 + 4 + \cdots$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Geometric sequences

[Sequences & Series · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

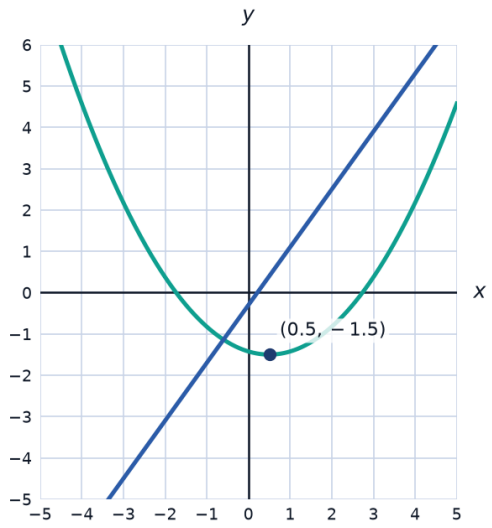
9 [Intermediate] • Target: Geometric series (finite)

Express the repeating decimal $0.3\overline{18} = 0.3181818\ldots$ as an irreducible fraction using the infinite geometric series formula.

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10 [Intermediate] • Target: Infinite geometric series

The second term of a geometric sequence is 12 and the fifth term is $\frac{32}{9}$. Find the common ratio r, the first term a_1, and the sum of the infinite series S_∞.



11 [Intermediate] • Target: Arithmetic sequences

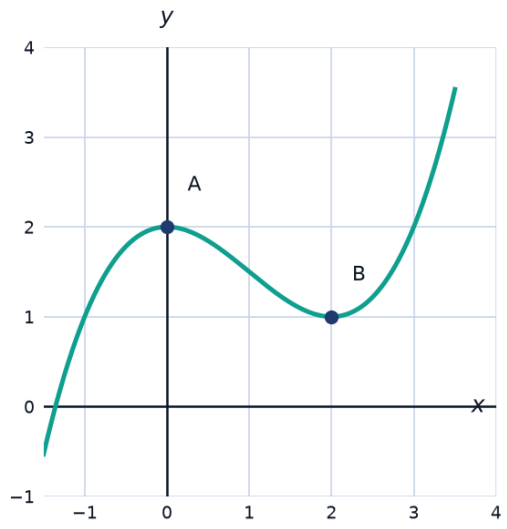
An infinite geometric series has first term $a_1 = 18$ and sum to infinity $S_\infty = 24$. (a) Find the common ratio r. (b) Calculate the third term a_3.

12 [Intermediate] • Target: Arithmetic series

An infinite geometric series has common ratio $r = -0.4$ and converges to an infinite sum of $S_\infty = 25$. Find the first term a_1, and compute the exact value of the partial sum S_4.

13 [Intermediate] • Target: Geometric sequences

Find the infinite sum of the geometric series $12 + 8 + \frac{16}{3} + \frac{32}{9} + \ldots$. Verify the condition for convergence.



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14 [Intermediate] • Target: Geometric series (finite)

In an arithmetic sequence, the 4th term is 17 and the 10th term is 41. Determine the common difference d, the first term a_1, and the partial sum of terms from a_{11} through a_{25} inclusive.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 1.01 Complex Numbers

Evaluate the exact value of the telescoping sum: $\sum_{k=1}^{50} \left(\frac{1}{k} - \frac{1}{k+1}\right)$ and evaluate the convergent infinite series: $\sum_{n=1}^{\infty} \frac{3}{4^n}$.

16 [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 3.01 Functions & Graphs

Evaluate the finite sum: $\sum_{k=1}^{6} 3(2)^{k-1}$

17 [Challenging] • Target: Arithmetic series • Cross-Lesson: 4.01 Matrices & Vectors

[Probability · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

18 [Challenging] • Target: Geometric sequences • Cross-Lesson: 5.01 Introductory Calculus

Find the 6th term in the expansion of $(3x - 2y^2)^8$ in fully simplified form.

19 [Challenging] • Target: Geometric series (finite) • Cross-Lesson: 6.01 Probability

Evaluate the sum of the convergent infinite series: $24 - 12 + 6 - 3 + \cdots$

20 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 1.01 Complex Numbers

Evaluate the finite geometric sum: $\sum_{k=1}^{8} 3 \cdot 2^{k-1}$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Arithmetic sequences

Find S_6 for the arithmetic sequence with $a_1 = -13$, $d = -1$.

a. -94

b. -89

c. -97

d. -93

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22 [Foundational] • Target: Arithmetic series

Find S_5 for the geometric sequence with $a_1 = -3$, $r = 3$.

a. -361

b. -363

c. -366

d. -365

23 [Foundational] • Target: Geometric sequences

Does the infinite geometric series with $a_1 = 6$ and $r = 1$ converge?

a. No — $|r| \geq 1$, it diverges.

b. Yes — it converges to $\frac{a_1}{1-r}$.

c. Yes — every geometric series converges.

d. Cannot be determined.

24 [Foundational] • Target: Geometric series (finite)

Consider a geometric sequence where the first term is $a_1 = \frac{1}{2}$ and the common ratio is r. If the sum of the first three terms is $S_3 = \frac{7}{2}$, find the value of the common ratio r.

a. 2

b. -2

c. 1/2

d. 3

25 [Foundational] • Target: Infinite geometric series

A geometric sequence has its first term a_1 and a common ratio r. The sum of the first three terms is 26, and the sum of the first six terms is 728. If $a_1 = 2$, find the value of r.

a. 2

b. 3

c. -2

d. -3

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26 [Foundational] • Target: Arithmetic sequences

A sequence is defined by the recurrence relation $a_{n+1} = 2a_n - 3$ for $n \geq 1$, with $a_1 = 5$. Consider a second sequence, b_n, where $b_n = a_n - k$ for some constant k. If the sequence b_n is a geometric sequence, find the value of k and the common ratio of the sequence b_n.

a. k = 3, r = 2

b. k = -3, r = 2

c. k = 2, r = 3

d. k = 3, r = 3

27 [Foundational] • Target: Arithmetic series

A geometric sequence has $a_1 = 6$ and common ratio $r = 2$. Find a_5.

a. 96

b. 192

c. 98

d. -96

28 [Foundational] • Target: Geometric sequences

Find the common ratio of the geometric sequence $5,\ 10,\ 20,\ 40,\ \ldots$

a. 2

b. -2

c. 3

d. 5

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Geometric series (finite)

Find the sum of the first 6 terms of the arithmetic sequence with $a_1 = 9$ and $d = 3$.

a. 99

b. 105

c. 144

d. 96

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30 [Intermediate] • Target: Infinite geometric series

An arithmetic sequence (a_n) and a geometric sequence (b_n) of non-zero real numbers share the same first term, so $a_1 = b_1$. The sequences also satisfy $a_5 = b_2$ and $a_6 = b_3$, with $b_2 eq b_1$. If the maximum value of the partial sum $S_N = \sum_{k=1}^{N} a_k$ of the arithmetic sequence is 153, what is the sum of the infinite geometric series $\sum_{n=1}^{\infty} b_n$?

a. 64

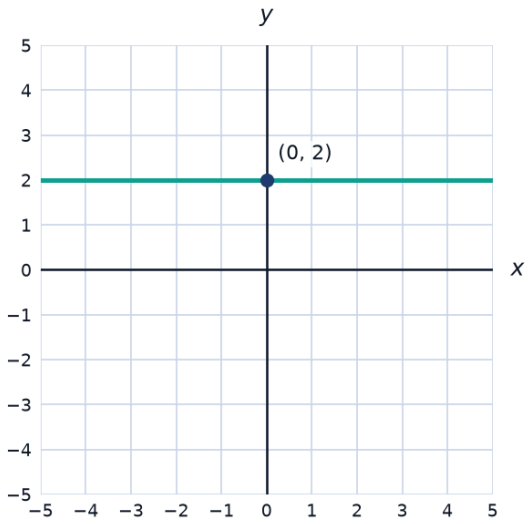
b. 96

c. 144

d. 192

31 [Intermediate] • Target: Arithmetic sequences

The graph of a linear function $f(x)$ is shown below. What is the value of $f(0)$?



a. -2

b. 0

c. 2

d. 4

32 [Intermediate] • Target: Arithmetic series

An infinite geometric sequence $a_1, a_2, a_3, \ldots$ of positive real numbers has common ratio $r < 1$ and sum S. The sub-sequence of terms whose indices are of the form $3k - 1$ for positive integers k (that is, $a_2, a_5, a_8, \ldots$) has sum $\sum_{n=1}^{\infty} a_{3k-1} = \frac{3}{13}S$. Furthermore, the infinite series formed by the squares of the terms has sum $\sum_{n=1}^{\infty} a_n^2 = \frac{9}{2}S$. What is the value of $\sum_{n=1}^{\infty} na_n$?

a. $\frac{27}{2}$

b. $\frac{81}{4}$

c. $\frac{9}{2}$

d. $\frac{27}{4}$

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33 **[Intermediate]** • **Target: Geometric sequences**

An infinite geometric sequence of positive real numbers has first term a_1 and common ratio $r \in (0, 1)$. Let $S_1 = \sum_{n=1}^{\infty} a_n$, $S_2 = \sum_{n=1}^{\infty} a_n^2$, and $S_3 = \sum_{n=1}^{\infty} a_n^3$. If $S_1 = 4$ and $S_2 = 8$, what is the value of S_3?

a. $\frac{256}{13}$

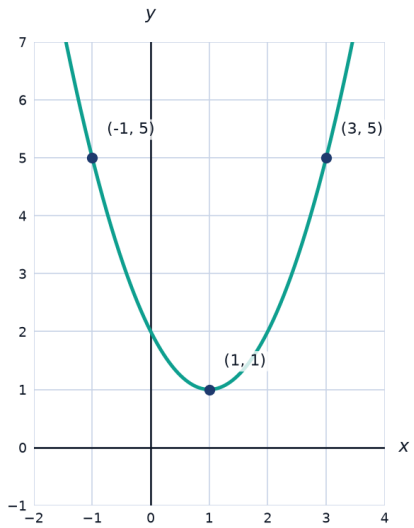
b. $\frac{128}{13}$

c. $\frac{64}{7}$

d. 16

34 **[Intermediate]** • **Target: Geometric series (finite)**

The graph of the function $y = f(x) = ax^2 + bx + c$ is shown below. The graph passes through the points $(-1, 5)$, $(1, 1)$, and $(3, 5)$. Determine the value of b.



a. 0

b. -2

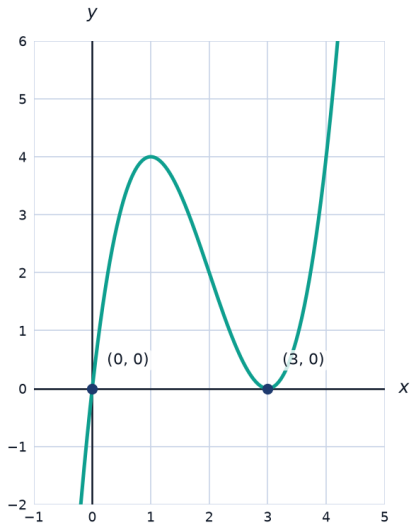
c. 2

d. 1

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35 **[Intermediate]** • **Target: Infinite geometric series**

Consider the function $f(x) = x^3 - 6x^2 + 9x$. Find the area of the region bounded by the graph of $f(x)$ and the x-axis.



a. 10

b. 27/2

c. 9

d. 27/4

36 **[Intermediate]** • **Target: Arithmetic sequences**

The first term of a geometric sequence is a. The common ratio is r. The sum of the first 3 terms is 21, and the sum of the first 6 terms is 189. If $r > 0$, find the value of a and r.

a. $a = 3, r = 2$

b. $a = 7, r = 1/2$

c. $a = 3, r = \sqrt{2}$

d. $a = 21, r = 1$

37 **[Intermediate]** • **Target: Arithmetic series**

The first three terms of a geometric sequence of positive real numbers are $k - 1$, $k + 2$, and $3k$, in that order. What is the value of the 6th term of this sequence?

a. 24

b. 48

c. 96

d. 192

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Geometric sequences • Cross-Lesson: 5.01 Introductory Calculus

The second term of a geometric sequence of positive real numbers is k, and the fifth term is $k^2 - 20$. If the common ratio of the sequence is 2, what is the sum of the first 6 terms of the sequence?

a. 155

b. 160

c. 315

d. 630

39 [Challenging] • Target: Geometric series (finite) • Cross-Lesson: 6.01 Probability

Consider the arithmetic sequence with first term $a_1 = 5$ and a common difference $d = 3$. What is the formula for the n-th term, a_n?

a. $a_n = 5 + 3(n - 1)$

b. $a_n = 3n + 2$

c. $a_n = 5n + 3$

d. $a_n = 5 + 3n$

40 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 1.01 Complex Numbers

An arithmetic sequence has its n-th term given by $a_n = 3n + k$, where k is a constant. If the sum of the first 5 terms of this sequence is 65, find the value of k.

a. 1

b. 2

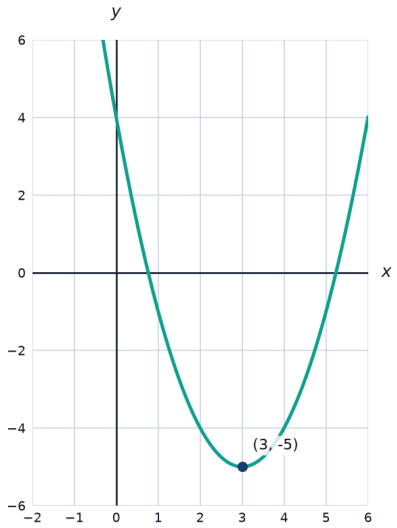
c. 3

d. 4

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41 [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 3.01 Functions & Graphs

The graph of the function $f(x) = x^2 - 6x + k$ is shown below. The graph has a vertex at $(3, -5)$. Determine the value of the parameter k.



a. 4

b. 1

c. 9

d. -4

42 [Challenging] • Target: Arithmetic series • Cross-Lesson: 4.01 Matrices & Vectors

In an arithmetic sequence, the 3rd term is $k + 7$, the 6th term is $3k - 1$, and the 11th term is $6k - 8$, where k is a real constant. What is the first term, a_1, of this sequence?

a. 6

b. 10

c. 19

d. 26

43 [Challenging] • Target: Geometric sequences • Cross-Lesson: 5.01 Introductory Calculus

A geometric sequence has its first term $a_1 = 5$ and its common ratio $r = 2$. If the sum of the first n terms of this sequence is $S_n = 315$, find the value of n.

a. 4

b. 5

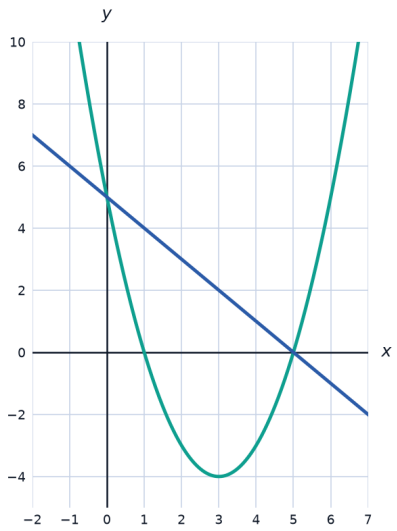
c. 6

d. 7

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44 [Challenging] • Target: Geometric series (finite) • Cross-Lesson: 6.01 Probability

Consider the functions $f(x) = x^2 - 6x + 5$ and $g(x) = -x + 5$. The graphs of these two functions intersect at two points. Let these points be A and B. A third function, $h(x)$, is defined as the difference between the x-coordinates of these intersection points, i.e., $h(x) = |x_A - x_B|$. What is the value of $h(x)$?



a. $\sqrt{16}$

b. $\sqrt{25}$

c. $\sqrt{36}$

d. $\sqrt{49}$

45 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 1.01 Complex Numbers

In an arithmetic sequence, the first term is $a_1 = 4$ and the fifth term is $a_5 = 28$. What is the value of the tenth term, a_{10}?

a. 52

b. 56

c. 58

d. 64

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